

Vladislav Vitmaer

CS4500 003

Operating Systems Project 4 Memory Management

We have neither given nor received unauthorized assistance on this work

Name of the VM: VladislavUbuntu

Password: ' ' The password is a single character the spacebar, ascii value of 0x32

Path: /home/vlad/class/project4

Description of Project:

The project was an exploration of memory mapping for the `task_struct`. We had to walk thru the maple tree and inspect some of the virtual address tables.

Pseudocode:

I iterate thru the `task_struct` list to find the one matching the PID argument.

I pass the PID relevant task struct to my helper function, which:

- gets a `read_lock`

- calls the `for_each_vma` macro

- executes some required math operations on it

Conclusion:

Memory management is a very complicated process. Even at the operating systems level here we are only touching a couple of APIs exposed to us like the `for_each` macros and the struct, which we don't really have direct access to since it could change. This project required use of the `read_lock` or the program would crash without it.

Lessons Learned:

Kernal debugging is hard. Seeing a segment fault was a good thing after hanging the vm.